Robot Behavior

الفصل الدراسي الأول

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Code	Chosen Course Name	P-R
CSE434	Selected Topics in Control	CSE224
CSE435	Robotics	CSE224

Assessment Method	Marks
Project	25
Quizzes	10
Midterm	15
Final	100
Total	150

PROJECT: TASK 1

- **Make a team group of (5-10) students.**
- **Choose one to be a leader.**
- **Try to find Idea for your project.**

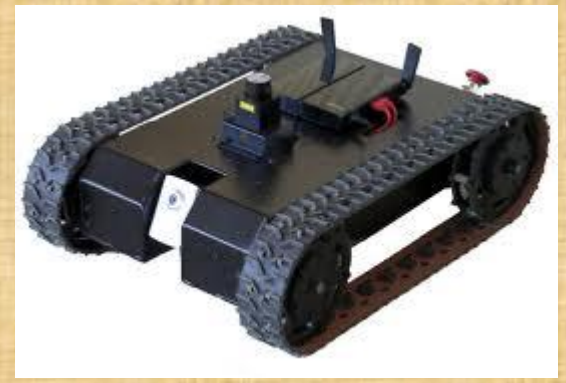
Agenda

- Activities of mobile robots.
- Activities of gantry robots.
- Feedback.
- Random activity.
- Subsumption.
- Input and output requirements.
- Distributed processing.

Mobile Robots

It is an automatic machine that is capable of movement in any given environment.











The activities of mobile robots

- Moving around
- Detecting and responding to light
- Proximity
- Contact
- Communication
- Navigation

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The activities of mobile robots

❑ Moving around:

- ➢ All mobile robots move from place to place.
- ➢ They need to be able to move forward, reverse, and to turn left or right.
- ➢ Variable speed is less important and often unnecessary.

❑ Detecting & Responding to light :

- ➢ One of the problems with using light sensors is that they may be confused by room lighting or sunlight.
- ➢ Pulsed light sources are one way out of this problem.

video

The activities of mobile robots

❑Proximity

- Another use for light is for *proximity sensing*.
- Proximity sensors tell the robot when it is near to, but not actually touching an object.
- It can also be used to measure the distance from a robot to some object.

- *Explain how light sensor detects a nearby object.*

1. A *light sensor* detects a nearby object by detecting the light reflected back from it.
2. If the intensity of the reflected light exceeds a certain level, the robot knows that something is there.
3. A *light detector* aimed sideways can be used to keep a robot at a fixed distance from a wall.

The activities of mobile robots

❑ Contact :

- *Explain how can the robot contact with an obstacle such as a fairly massive object or a wall.*

1. The robot has *bumpers* or possibly wiry 'antennae' arranged so that they are touched when the robot runs into anything.
2. The usual response is to reverse a short distance, turn slightly to left or right, then move forward to try again.
3. If the robot has a pair of bumpers, at front left and right, it is possible for the robot to work out which is the best direction to turn.

video

The activities of mobile robots

❑ Communication :

- Most robots need to interact with humans, and those programmed to play games interact more than most.

video

Multicommunication between robotics.

video

The activities of mobile robots

❑ Navigation :

- Given that a robot is mobile, it seems reasonable for it to know where it is.
- There are basic methods of navigation, such as

1) Wall -Following

a common type of robot behavior it is often used by maze solving robots. **video**

2) Line Following

- ✓ Line following is a special form of contact behavior
- ✓ The robot stays in contact with a line painted on the surface over which it is moving.
- ✓ It is one of the most reliable techniques for guiding a robot from one place to another.

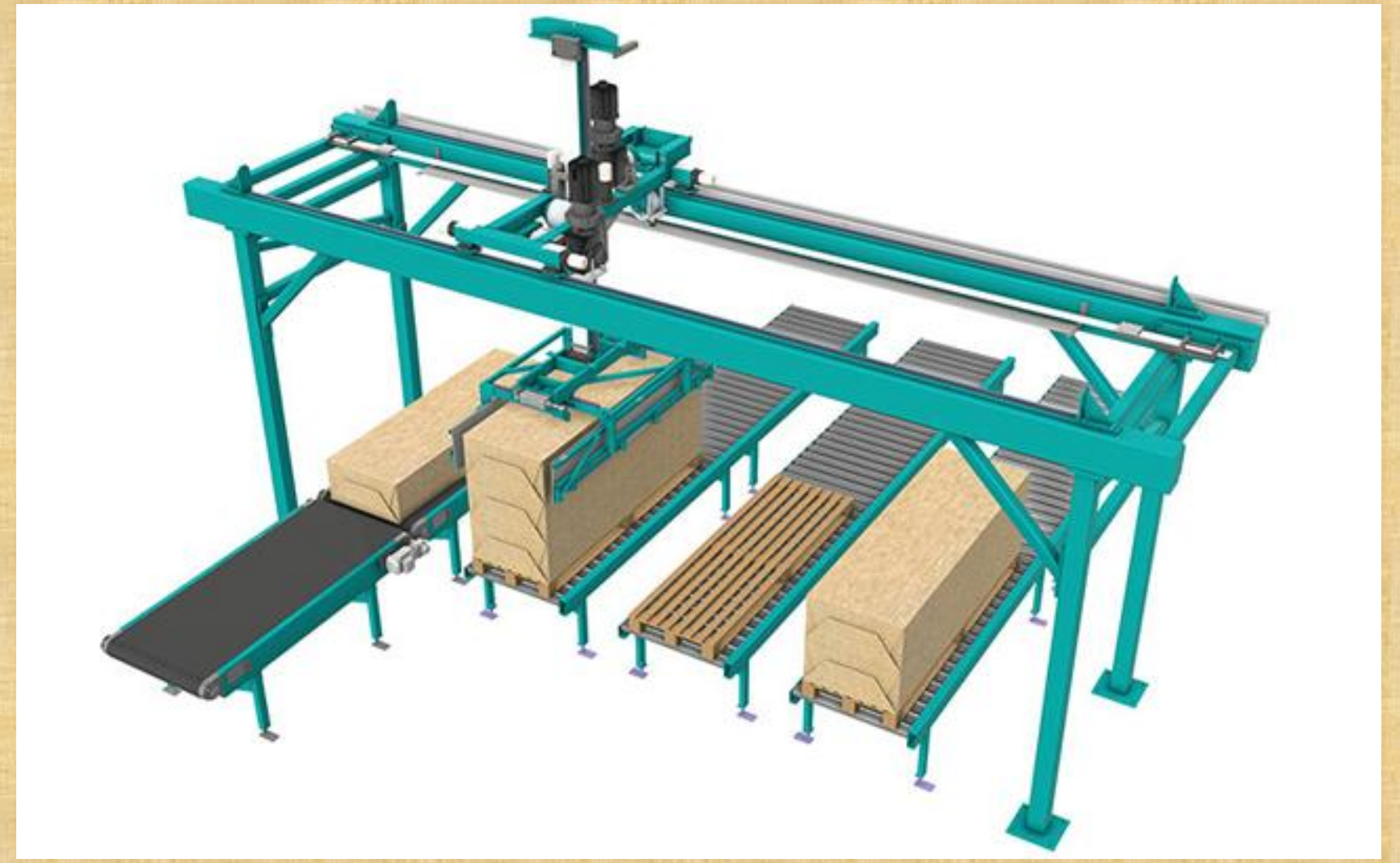
3) Homing on a light source

The activities of mobile robots

❑ Navigation :

- Some operations require the robot to move around in ‘free space’, without reference to lines, walls or beacons.
- It might be thought that switching on the drive motors for precisely controlled periods would give good positional information.
- This does not work in practice. For one thing, the two drive motors do not run at exactly the same speed
- When turning, it is not possible to control the turning angle precisely.
- Errors of this kind are cumulative and it is not long before the robot completely loses its bearings
- We can counteract the differences between motors in several ways (tachometer or stepper motors)
- However, even these may not entirely solve the problem. Depending on the nature of the surface and of the tyres,
- there is inevitably a small amount of slipping

Gantry robot



Activities of gantry robots

- **A gantry robot :**
  - ✓ Operates over a clearly defined rectangular area.
  - ✓ It picks up objects from any point in the area and sets them down at another point in the area.
- The tool (often a gripper) is suspended from a small trolley-like frame, and can be lowered and raised.
- The frame has wheels and runs on a pair of rails so that it can travel from one side of the area to the opposite side.
- Like mobile robots, gantries can be programmed to solve mazes. But mobile robots are apt to lose their bearings.
- Because the travelling frames can be precisely positioned by keeping track of their x and y coordinates, a gantry robot can never lose its bearings.
- Gantry robots are used in industry when very heavy loads are to be handled

Video video

Feedback

- As an example of feedback, take an ordinary refrigerator.
- When the temperature inside it rises above a given level the refrigerator pump is turned on automatically.
- It stays on until the temperature has fallen to a given level.
- In this way the temperature inside the refrigerator is held within close limits.
- This is called negative feedback because an increase in temperature results in the interior being cooled down
- **How can feedback be used in robotics activities?**
  - When the robot veers toward the wall, the amount of reflected IR increases.
  - The sensor detects this increase.
  - The program responds to this increase by steering the robot to the right, making it veer away from the wall.
  - The reverse happens as the robot veers away to the right.

video

Feedback

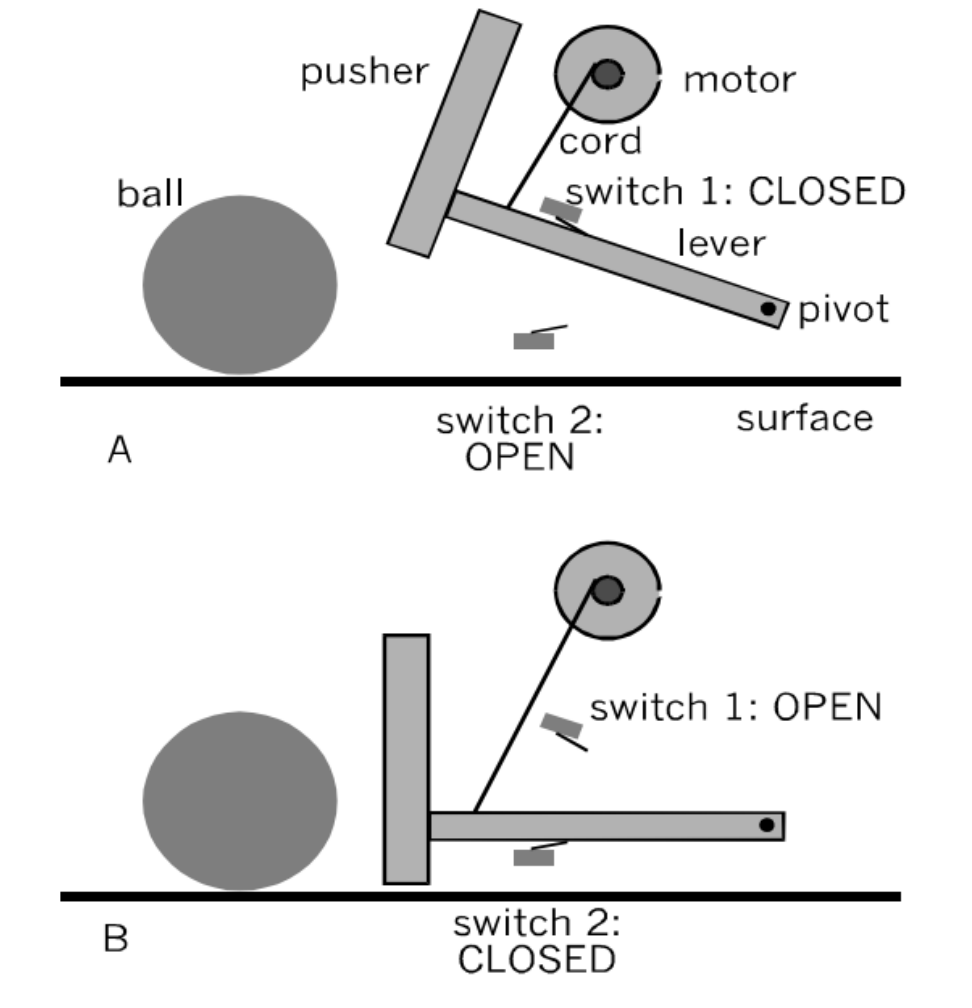
- The usual function of negative feedback is to hold things constant.
- It produces stability.
- There is also positive feedback.
- In the case of the wall-following robot described above, suppose that by mistake the output lines to the motors were swapped.
- Then the slightest deviation from the correct robot–wall distance would cause dramatic results.
- The robot would either veer permanently away from the wall or crash into it.
- Positive feedback results in instability, which is something to be avoided in robot behavior.

video

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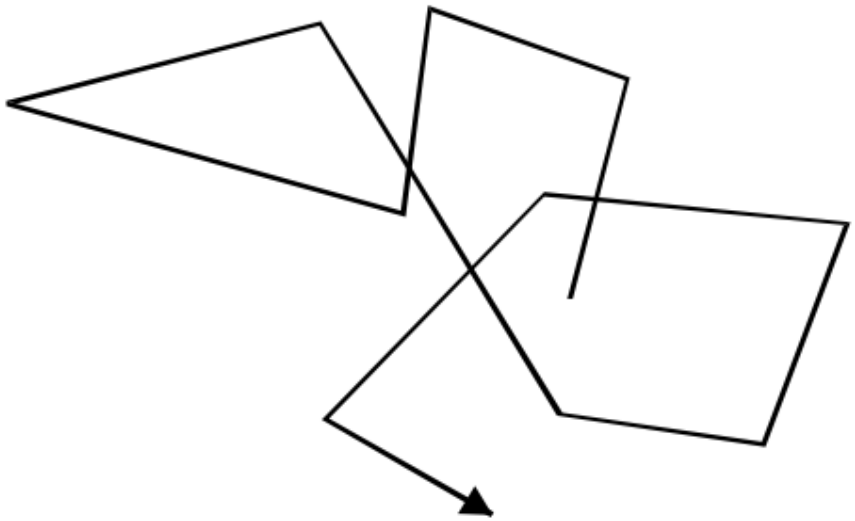
Feedback

- There is another type of feedback that is virtually essential in a robot system.
- This type of feedback is based on *limit switches* video



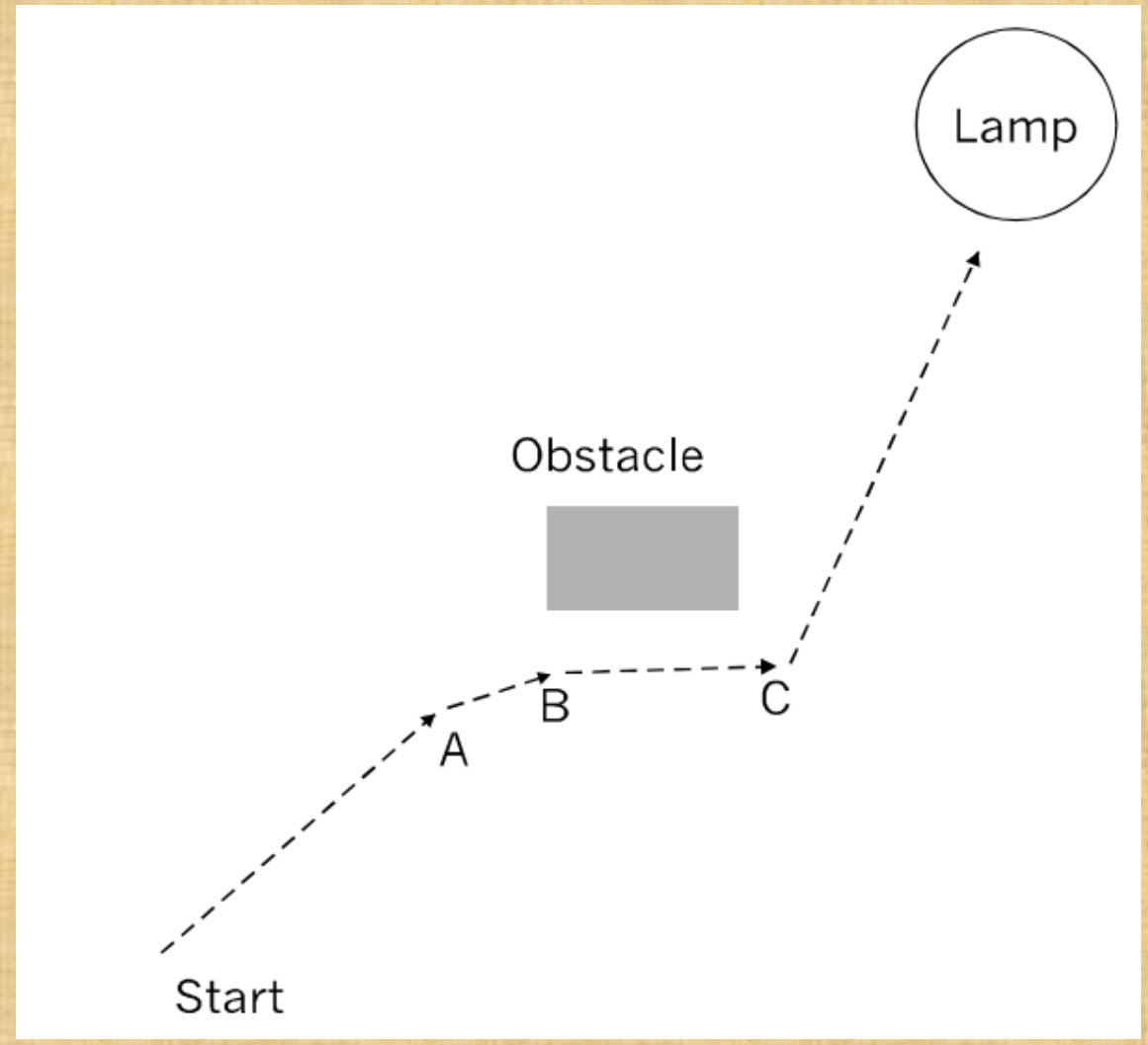
Random activity

- If a robot is programmed to run f
  distance, then turn through a ra
  indefinitely in the same routine i
- We say that it is literally perform
  - Random behavior is produced i
    number generation routine. Act
    random, but a predictable serie
    long interval that it appears to
    random.
  - Randomness has other and mo
  - A robot that solves a maze may
    choice whenever it has to go either left or right at a junction.
  - If it is also programmed to remember which choices it made at each junction and which choices took it successfully to its goal, it can eventually learn to run the maze correctly.
  - This is the basis of learning by **trial and error. The same type of learning technique can** be applied to other learning tasks. Video video



A Monte Carlo Walk usually results in staying in more-or-less the same place.

Subsumption



Subsumption

- When mobile robot reaches point A, its proximity detector detects an obstacle between it and the source of light, It immediately stops its homing behavior.
- Instead, it enters a phase of avoidance behavior.
- It turns and proceeds to B. There, it still detects the obstacle, so turns more, heading for C.
- It has avoided the obstacle and, once it is at C there is a clear path to the lamp. It resumes its homing behavior.

Video

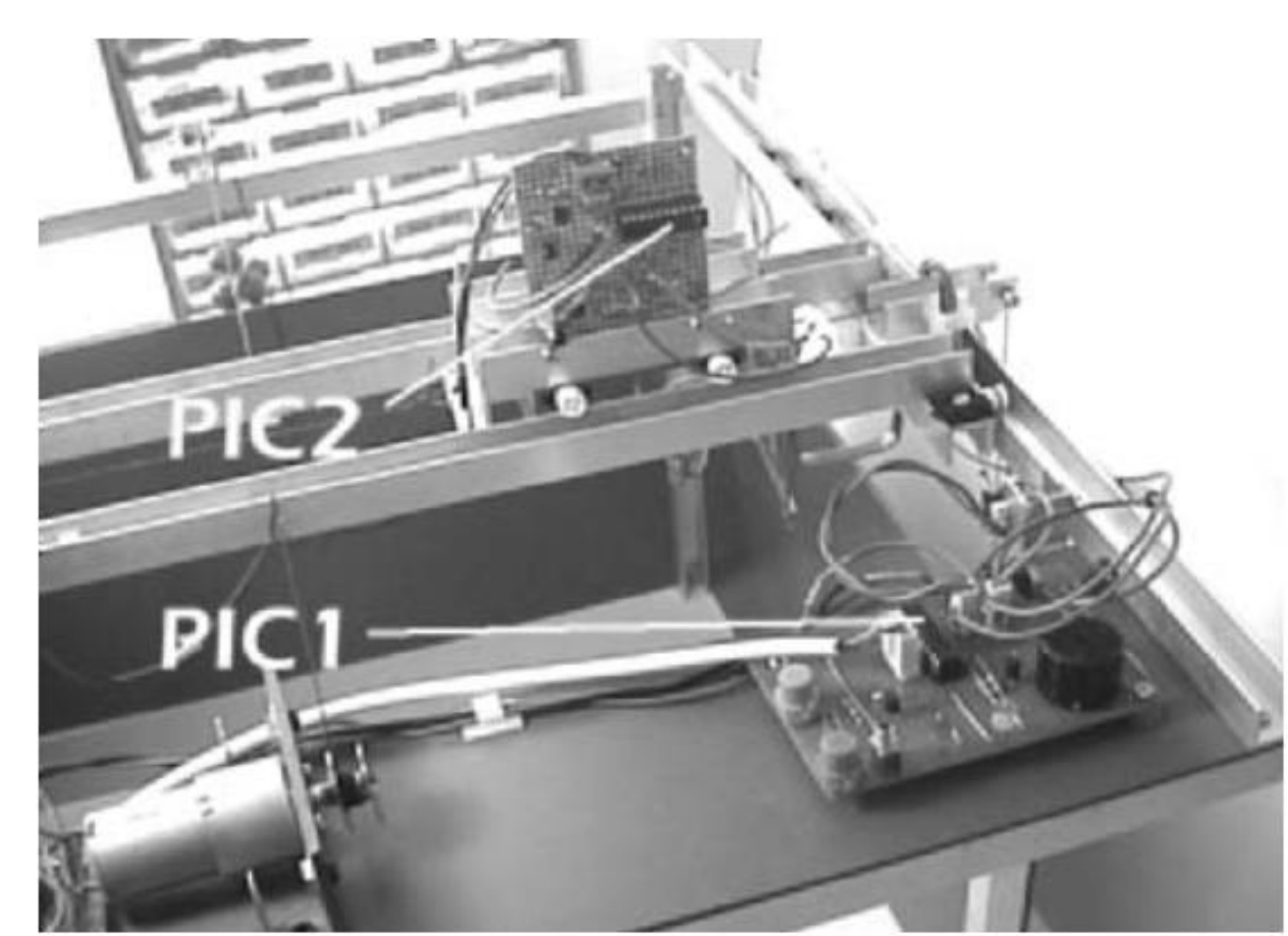
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Distributed processing

- Programming simultaneous activities on a single controller is possible but difficult.
- One solution is split the tasks between two or even more controllers.
- Each runs independently, except for occasional handshaking signals sent from one to the other to tell it what it is doing.
- This is known as distributed processing.
- An example of distributed processing is the *Horseshoe game running on the Gantry.* Video

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Distributed processing



The Gantry robot (Project 6.5, p. 297)
uses two controllers for its more complicated activities. PIC1 is on the main frame. PIC2 is on the x-frame.

Questions

1. What are the activities of mobile robots? activities
2. How can feedback be used in robotics activities ?give an example . answer
3. Explain and show the subsumption behavior explain show
4. What is the meaning of Distributed processing ? give an example. answer